UCS760: Edge AI and Robotics: Reinforcement Learning & Conversational AI

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Course Objective: This course will provide students with introduction to the basic mathematical foundations of Reinforcement Learning for building real world computer vision applications, and Conversational AI for developing Chatbots.

Syllabus

Introduction: GPU Computing, Implementing Behaviours of Robots such as Manipulation, and Task Learning, Fundamentals of Reinforcement Learning for Vision and Deploying Conversational AI pipelines in JetsonI.

Manipulation: Overview of Manipulation in Robotics, Inverse Kinematics and Control, Gripping & Task Learning.

Reinforcement Learning: Introduction to RL: RL agents, Dynamic Programming, Monte Carlo's and Temporal-Difference Methods, OpenAI Gym, RL in Continuous Spaces.{Added Lectures, Summaries}

Conversational AI (NLP): Natural Language Processing: Introduction to NLP, BERT, Megatron, Applications of NLP: Information Retrieval, Intent Slot Filling, Machine Translation, Punctuation & Capitalization, Question and Answering Machine Machine, Relation Extraction, Sentiment Analysis, Token Classification in NeMo.

Conversational AI (Speech Processing): Automated Speech Recognition: Introduction to ASR, Architectures: Jasper/QuartzNet/CitriNet, Text to Speech: TTS-Tacotron2/WaveGlow and Jarvis Deployment

Laboratory Work:

- Manipulation Lab: Building Pick-n-place.
- Manipulation Lab: Object Assembly.
- Game Agent: Open AI Gym (Jetbot in simulation).
- Conversational AI VoiceBot: Verbal JetBot commands/feedback, ect (optional mic/speaker).

Course Learning Objectives (CLO)

After the completion of the course the student will be able to:

1. Introduce the advanced fundamental problems of reinforcement learning and conversational AI.

2.Working with various simulation environments for deployment of computer vision model.

3.Provide understanding of approaches, concepts and algorithms used in reinforcement learning and conversational AI with practical exercises.

4.Utilize programming and AI training & deployment tools for relevant model building in both edge hardware devices and simulation environments.

Text Books

1. Wiering, Marco, and Martijn Van Otterlo. "Reinforcement learning." Adaptation, learning, and optimization 12 (2012): 3.

2.Russell, Stuart J., and Peter Norvig. "Artificial intelligence: a modern approach."Pearson Education Limited, 2016.

3.Jurafsky, Dan and Martin, James, Speech and Language Processing, Second Edition, Prentice Hall, 2008.
4.Daniel Jurafsky and James H. Martin, "Speech and Language Processing", 3rd edition draft, 2019 [JM-2019].

Reference Books:

1. Goodfellow, Ian, Yoshua Bengio, and Aaron Courville. "Deep learning." MIT press, 2016.

2.Mark Gales and Steve Young, The application of hidden Markov models in speech recognition, Foundations and Trends in Signal Processing, 1(3):195-304, 2008.

3.Daniel Jurafsky and James H. Martin. 2009. Speech and Language Processing: An Introduction to Natural Language Processing, Speech Recognition, and Computational Linguistics. 2nd edition. Prentice-Hall.

4."Reinforcement Learning: An Introduction" by Richard S. Sutton and Andrew G. Barto: https://webdocs.cs.ualberta.ca/~sutton/book/the-book-2nd.html

5.David Silver's course: http://www0.cs.ucl.ac.uk/staff/d.silver/web/Teaching.html

6."Deep Reinforcement Learning: Pong from Pixels" by Andrej Karpathy: https://karpathy.github.io/2016/05/31/rl/

7.Talks on Deep Reinforcement Learning by John Schulman: https://www.youtube.com/watch?v=aUrX-rP_ss4 , and his Deep Reinforcement Learning course http://rll.berkeley.edu/deeprlcourse/